# PERPETUAL BITCOIN – Whitepaper V1.1

## Immutable. Permissionless. Un-cheatable.

### 1. Abstract

**Perpetual Bitcoin (PB)** is a finished revolutionary fully immutable, trustless, fractional price-based unlocking protocol designed to correct the structural failures of modern crypto.

PB eliminates admin keys, time-based vesting, transferability exploits, liquidity rugs, governance capture, and user/whale sell pressure.

All **21,000,000 PB** are minted at genesis and placed into a single global Vault. Only a small fraction enters circulation at launch through presale allocations and initial liquidity. The remaining supply is released only when the market earns it, through price-based unlocks.

Every PB purchase mints a non-transferable receipt tracker NFT (PBt) that records the buyer's price, position, and next unlock point.

Unlocks occur only when the market reaches the next 1.5555× price level for each purchase, NOT by time, NOT by insiders, NOT by human discretion.

PB is a mathematical constitution: One Vault, one supply, one set of rules, zero exceptions.

### 2. The Problem with Crypto

Crypto was meant to be trustless, immutable.

> Instead, it became: - admin controlled - upgradeable - pausable - manipulable - insider weighted - vesting cliff driven - liquidity rug prone - SELLING pressure driven

**PB identifies three structural failures:**

#### 2.1 Human Control

Admin keys, multi sigs, governance votes, and upgrade proxies allow insiders to change rules, mint tokens, freeze wallets, or drain liquidity.

#### 2.2 Time Based Vesting

Unlock cliffs create predictable dump windows that bots exploit and insiders front-run.

Time vesting is the most destructive mechanic in token launches. => predictable selling pressure

#### 2.3 Transferability of Locked Positions

Founders and insiders routinely sell their vesting bags OTC, transferring obligations to unsuspecting buyers. This is the hidden rug no one talks about.

## PB exists to eliminate all three.

## 3. PB Philosophy

PB is built on four non-negotiable principles:

### 3.1 Immutability

No admin. No owner. No upgrades. No backdoors. No emergency pause.

### 3.2 Fairness

No VC allocations.  No special time unlock rules. No blacklist function.

### 3.3 Price Based Unlocks

Unlocks must be earned, not scheduled.

PB unlocks only when the market reaches the next 1.5555× (trigger) level for your purchase.

### 3.4 Non-Transferability for Perpetual Bitcoin Claim (PBc) and  PB tracker PBt

ERC-20 Claim token (PBc) and ERC-721 Tracker NFT (PBt) Positions cannot be sold, transferred, or dumped.  These are bound to your wallet.

**Perpetual Bitcoin is not a token.**

**PB is a revolutionary system.**

**Fractional Value Release.**

## 4. System Overview

PB consists of four immutable components:

### 4.1 Perpetual Bitcoin PB token ERC20

21,000,000 PB minted at genesis. In the vault.

3.69% of any purchase is liquid and freely tradable.

### 4.2 Perpetual Bitcoin claim PBc (Locked PB) ERC20

21,000,000 PBc minted at genesis. In the vault.

96.31% of every purchase becomes PBc token via the Vault, locked PB tracked inside the Vault.

PBc cannot be moved, transferred, or sold directly! Vault is in control via contract **1 to 1 with PB**

### 4.3 The Vault

A single global contract holding all 21,000,000 PB and 21,000,000 PBc at genesis.

The Vault:

- Keeps track of a cumulative counter
- assigns PB + PBc to presale buyers
- routes swaps PB->PBc for regular purchases
- assigns PB + USDL to LP when conditions are met (distribution phase)
- executes tranches – unlocks and sells PB to AMM, send USDL to BPt holder – updates PBt
- updates users PBt when user enters/activate a recovery / inheritance address
- mints 1 PBt NFT for each purchased batch. Burns the PBt when the last 10 Satoshi is sold
- mints PBr /PBi NFT's. burns these when activated.
- cannot be modified

The Vault is the only source of PB entering circulation.

**4.4 Tracking NFTs (PBt)** user's on-chain contract with the Vault.

Each purchase the Vault mints a non-transferable ERC-721 Tracker NFT PBt containing:

- buy price
- PB amount
- PBc locked amount
- mint block
- holders address
- full history + next unlock point
- recovery payout address + inheritance address (available via frontend) *

## 5. Buy Flow & Liquidity Mechanics

Every buy follows the same immutable process:

### 5.1 User Allocation/buy

The buyer receives 100% of their PB allocation/buy, consisting of:

- 3.69% PB delivered liquid freely tradable.
- 96.31% in PBc representing their locked portion PB inside the Vault. Nontransferable.
- 1 PBt NFT per purchase. Nontransferable.

### 5.2 Value Flow (Distribution Phase Only)

This phase is active only while the Vault holds more PB than the total outstanding PBc supply.

Every buy activates PB's Convergent Netting Engine, a three-layer system that settles unlocks internally, grows liquidity, and minimizes AMM impact:

****Layer 1: Unlock Netting (zero AMM impact):****

The Vault scans all PBt positions eligible to unlock at the current price. Instead of buying PB on the AMM and then selling it back for unlocks, the buyer's USDL is routed directly to unlock holders, and the corresponding PB stays in the Vault credited to the buyer. Settlement price: each PBt's immutable `nextTriggerPrice`, not spot.

****Direction detection determines the remaining flow:****

****NET BUY**** (buyer USDL > total unlock USDL):

- 36.9% of the net USDL is paired with Vault excess PB and added to the LP (no selling, buyer's USDL paired directly with Vault PB)
- 63.1% of the net USDL executes a single AMM buy (pushes price UP)
- The vault uses 2× PB for the LP contribution: half is paired with USDL in the LP, half is given to the buyer as compensation (liquid PB)
- Result: 1 AMM swap + 1 addLiquidity per buy (vs. old: 1 big buy + up to 25 sells + 1 LP sell + 1 addLiquidity)

****NET SELL**** (total unlock USDL > buyer USDL):

- All buyer USDL settles unlocks internally
- Remaining unsettled unlocks are batched into a single AMM sell
- No LP contribution (no excess USDL available)
- Result: 1 AMM swap per buy

The Vault uses a convergent binary search to find the maximum number of unlocks (K) where the post-netting AMM price still reaches all K trigger prices. This guarantees mathematical consistency, no unlock is settled unless the market actually supports it.

No PB is ever minted during this process. All PB used for LP pairing or buyer compensation is drawn from the pre-minted 21M supply held in the Vault.

**Vault PB Balance > Total PBc Outstanding (21M - Vault PBc balance)**

This ensures the Vault always retains enough PB to fulfill all future unlocks and redemptions.

### 5.3 Value Flow (Final Phase)

The system enters the Final Phase when the Vault PB balance is no longer strictly greater than the total outstanding PBc supply.

At this point:

- The 75% LP contribution rule is permanently disabled
- No more PB is assigned to LP under any condition
- No PB leaves the Vault except through trigger-based unlocks (1.5555× price events tied to PBt schedules).
- The Vault transitions into a **fixed-supply, unlock-only system**, where every remaining PB is reserved exclusively for fulfilling:
  - PBc redemptions (1:1 PB backing)
  - PBt unlock tranches (trigger-based sales)

This transition is **automatic, irreversible, and enforced by immutable contract logic**.

It guarantees:

- **Full collateralization** of all outstanding PBc tokens
- **No over-allocation** of PB under any circumstance
- **Mathematical integrity** of the system through the final unlock

### 5.4 VLock (VoluntaryLock) Voluntary Locking and LP Bonus

VLock is an opt-in mechanism for voluntarily converting liquid `PB` into `PBc` while minting a `PBt` receipt. To encourage participation the Vault may pay an immediate bonus funded strictly from harvested LP proceeds. Key policy:

Purpose

- Provide an immutable, keeperless incentive that rewards participation without exposing Vault reserves.

On-chain flow (summary)

- User calls `voluntaryLock(amount)` on the `Vault` (user must `approve` `PB`).

- Vault snapshots a TWAP price (1-minute) for the entry, mints a `PBt`, converts the submitted `PB` to `PBc`, and updates balances.

- Vault performs an on-tx harvest (`_harvestLPRewards()`), crediting two fee pools: `accumulatedFeesUSDL` and `accumulatedFeesPB`.

- If the USDL-equivalent of the locked amount >= `MIN_VLOCK_BONUS_USDL` the caller receives a fixed share of the *remaining* fee pools:

  `usdlAllocation = accumulatedFeesUSDL * VLOCK_BONUS_PCT / PCT_DENOM`

  `pbAllocation  = accumulatedFeesPB * VLOCK_BONUS_PCT / PCT_DENOM`

- Allocations are deducted from the accumulators and paid (USDL transfer + PB in-kind transfer). State updates follow checks → effects → interactions and execute under `nonReentrant`.

Eligibility & pricing

- Gate: USDL only. `MIN_VLOCK_BONUS_USDL = 100 USDL` (protocol default). Use PulseX 1-minute TWAP for conversion; if TWAP is unavailable use the last on-chain pair price only for conversion.

- Payout: fixed share of remaining pools (`VLOCK_BONUS_PCT` of each accumulator) this implements geometric decay (pools shrink by the same factor each eligible call) and prevents single-lock percentile dependence.

Economics & safety

- `VLOCK_BONUS_PCT` default = 5.555% (5555 / 100000). Allocations use the pool-percent formula above; do not compute bonuses as a percent of the user's lock.

- PB allocations are paid in-kind and must decrement both `accumulatedFeesPB` and `vaultPBBalance` when executed.

- Do not `require (allocation > 0)` , `voluntaryLock` must succeed even if allocations are zero.

Observability & events

- Emit `LPRewardsHarvested(usdl,pb)`, `VLockBonusPaid(user,usdl,pb)`, and `VLockExecuted(...)` for auditing. Events include `PBt` id, locked amount, USDL_equivalent, allocations paid, and pre/post accumulator values.

Rationale

- Releasing a fixed share of the remaining pools per eligible call avoids rewarding large lockers disproportionately and ensures the reward budget decays predictably over time. This aligns incentives for broad participation while keeping payouts strictly funded by harvested LP rewards.

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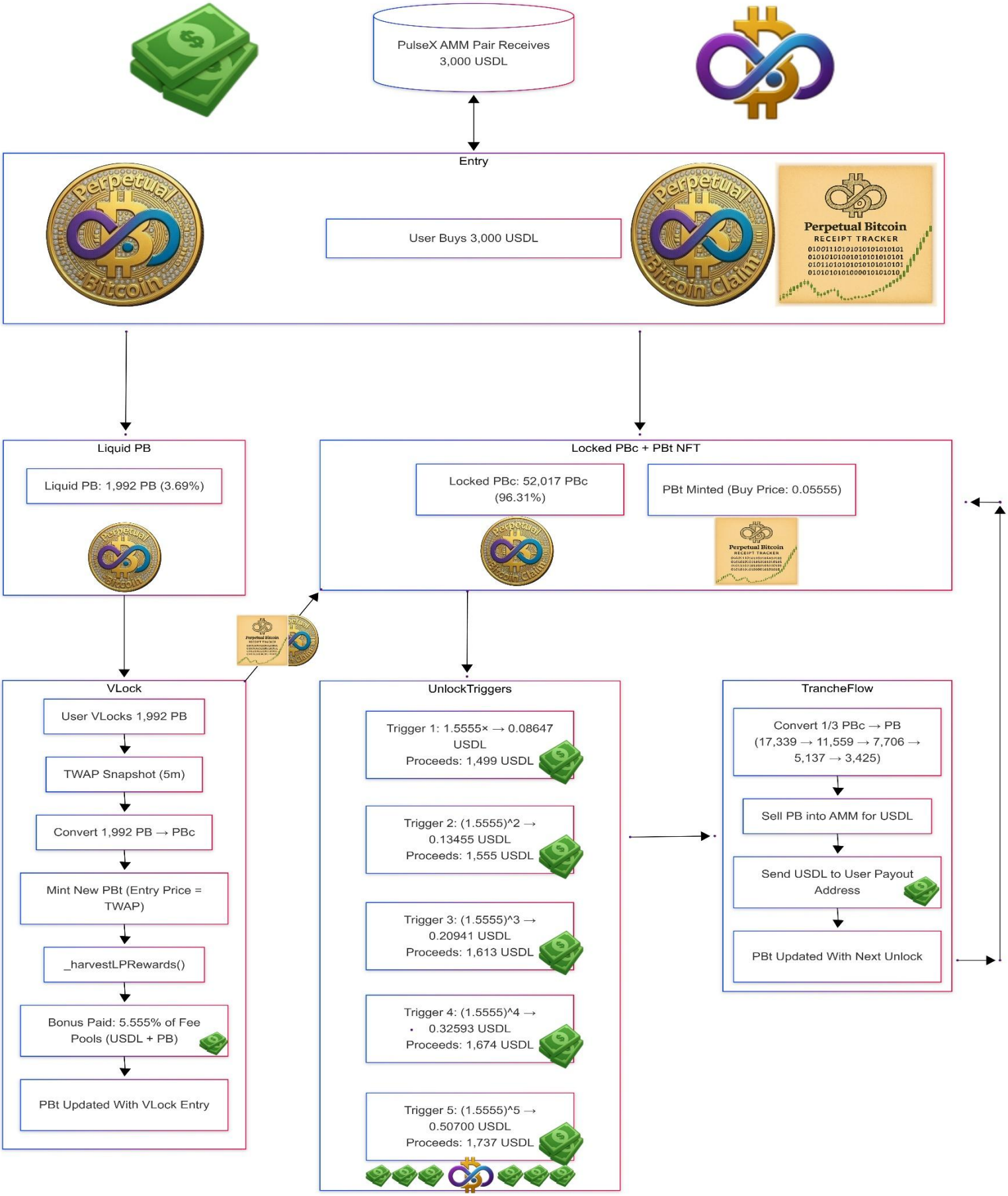

## 6. Presale Architecture

PB will have MAX 555 presale blocks @ $555.5 each:

### 6.1 Founder, DEV, OA, Advertisement Allocation (1 + all PRIME locations)

****1.25 M PB = ~6% total supply****

102 founder spots are allocated to prime positions: position 1 + first 101 primes ≤547 (2, 3, 5, 7, 11, ..., 547).

These blocks are allocated to Founder, DEV, Operational, and Advertisement Addresses.

- No buy-in required
- ***Bound to the same 3.69% / 96.31% rule***
- ***Same unlock progression - Same non-transferability - Same payout rules - No privileges***

### 6.2 Public Presale Spots (#38–555)

The remaining 453 blocks are available to the public @ 555.5 USDL (Max 5 blocks per address)

Each paid spot receives a Receipt NFT 'PreSalePBIOU' for variable PB when entering: PB = 555.5 / entry_price (10,379-15,054 PB). At launch the pre-sale contract will convert this IOU to:

- Variable PB total (PB + PBc)
- 3.69% liquid PB
- 96.31% locked PBc
- 1 PBt NFT with next unlock point

### 6.3 Rising Presale Price Curve

PB allocation varies by block: PB = 555.5 / entry price. Each block records its own entry price (used for unlock triggers):

- Spot #1: 0.0369 USDL per PB → 15,054 PB allocation
- Spot #555: 0.05352 USDL per PB → 10,379 PB allocation
- Target launch Price: 0.05555 USDL per PB

This ensures:

- early believers are rewarded earlier
- late presale buyers approach launch price
- no cliffs - no insider advantage - no special deals

**7. Launch Liquidity Goal (PulseX)**

The sets the launch price will be : 0.05555 USDL per PB

Liquidity is permanent. The Vault never removes LP tokens. Liquidity only grows during the distribution phase through:

- NET BUY netting: 36.9% of net USDL paired with Vault excess PB directly into LP (zero sell pressure)

LP growth uses buyer's USDL paired directly with Vault's excess PB, no PB is ever sold to create LP.

This means LP contributions create zero downward price pressure.

Once circulating supply reaches 21M, LP growth becomes purely market-driven.

**8. Unlock Mechanics**

PB unlocks only when there is marker demand! And uses a 1.5555× geometric unlock multiplier.

**8.1 Unlock Trigger in EACH PBt**

Each time the market price reaches the next 1.5555× trigger level in a PBt:

- the Vault swaps 1/3 of the users remaining PBc (of that receipt tracker) to PB
- During a buy: settled internally via netting (buyer's USDL → holder, no AMM sell)
- Standalone: PB sold to AMM for USDL, proceeds sent to holder
- Sends ****100% of proceeds**** to the user's payout address

Note: The 36.9% LP rule applies only to the NET BUY portion of buys, not to trigger-based unlocks. Trigger proceeds go entirely to the user.

**8.2 Why 1.5555×**

The multiplier (FIXED forever):

- absorbs AMM slippage while end goal is 1.5x + gas costs
- avoids borderline triggers and reduces tranche spam
- creates meaningful unlock spacing
- aligns with PB's symbolic numerology

**8.3 Geometric Decay**

Unlocks follow: Remaining PBc = $(^2/_3)^n$

This creates: - predictable decay, long-term alignment, no sudden sell pressure cliffs

## 9. Security Model

PB is designed to be uncheatable.

### 9.1 No Admin

No one can pause, upgrade, mint, drain, modify, override, kill the protocol.

### 9.2 Non-Transferability

Tracker BPt NFTs cannot be sold/transferred.

PBc cannot be sold/moved.

Positions cannot be sold.

### 9.3 Dual-Signature Payout Update, Recovery & Inheritance Options

After a buy is completed, the PBt ERC-721 holder may optionally set a recovery address and/or an inheritance address + password. Both can be set at any time after the buy, but only once each.

Once defined, they are immutable and cannot be changed.

When set, the vault protocol mints a non-transferable ERC-1155 PBr (recovery) or PBi (inheritance) NFT to the designated address.

To activate a payout switch, the receiver must present this PBr / PBi ERC-1155 NFT together with the password/PIN defined by the original holder at creation of the BPr or PBi.

Inheritance address overrides recovery address. If both activated, inheritance takes precedence.

This ensures proceeds recovery and succession planning without human intervention, admin control, or governance capture.

### 9.4 No Governance

No votes.

No committees.

No whales deciding rules.

PB is governed by math, not people.

## 10. Mathematical Appendix

### 10.1 Total Supply

21,000,000 PB + 21,000,000 PBc minted at genesis. **1PB=1PBc ALWAYS**

All tokens are stored in the Vault.

No further minting possible. Supply is permanently fixed.

### 10.2 Circulating Supply Definition

Circulating PB is defined as the total PB that has exited the Vault, including:

- Liquid PB held by users (3.69% of each buy)
- Locked PB represented by PBc held by users (96.31% of each buy)
- PB paired in liquidity pools (Vault-owned LP tokens)

Circulating PB = User Liquid PB + User PBc + PB in L

### 10.3 Distribution Phase → Final Phase

The system operates in two distinct phases:

#### 10.3.1 Distribution Phase

Active while the Vault holds more PB than the total outstanding PBc supply.

Vault PB Balance >Total PBc Outstanding

During this phase, the Vault contributes to LP using the 36.9% rule: 36.9% of net USDL from each NET BUY is paired with Vault excess PB directly into LP (zero sell pressure). The remaining 63.1% executes a single AMM buy.

#### 10.3.2 Final Phase

The system enters the **Final Phase** when the Vault PB balance is **no longer strictly greater** than the total outstanding PBc supply:

Vault PB Balance ≤ Total PBc Outstanding

At this point:

The 75% LP contribution rule is permanently disabled. No further PB is added to LP by the protocol. All remaining PB is reserved exclusively for fulfilling:

PBc redemptions (**1:1 PB backing**)

PBt unlock tranches (trigger-based sales)

This transition is automatic and irreversible, enforced by immutable contract logic. It guarantees that every PBc token remains fully redeemable, and that no PB is ever over-allocated.

### 10.4 Unlock Levels

Each PBt defines a geometric unlock progression based on the buyer's entry price:

$P_0$=Buy Price

$P_n = P_0 \times 1.5555^n$

Where:

$P_0$ is the original purchase price

$P_n$ is the price required to trigger the $n^{\text{th}}$ unlock

### 10.5 Tranche Decay

Each unlock tranche releases one-third of the remaining PBc balance in the PBt:

$T_n = (1/3) \cdot (2/3)^{n-1}$

Where:

$T_n$ is the fraction of the original PBr released at the $n^{\text{th}}$ trigger

This creates a **smooth geometric decay**, ensuring:

No cliffs

No abrupt unlocks

Long-term alignment between conviction and unlock velocity

**11. Philosophy & Legacy**

PB is not a token.  PB is a correction. A system that cannot be corrupted, compromised, or captured.

It exists because crypto lost its way, and someone needed to build the protocol that actually lives by the values everyone else only markets.

PB stands for:

- sovereignty
- immutability
- fairness
- transparency
- long-term alignment
- mathematical truth

## PB is designed to outlive our generation.

**Recovery** and **Inheritance** mechanics ensure PB positions can survive hacked wallets, lost keys, and even generational succession.

PB is the counterexample, the proof that crypto can be better.

**12. Conclusion**

Perpetual Bitcoin BP is a new class of protocol:

price-based, immutable, uncheatable, and perfectly fair.

It removes every vector of manipulation:

no admin, no upgrades, no vesting cliffs, no transferability, no liquidity rugs, no insider dumping.

PBc unlocks from a single AMM pair using market price from PulseX.

No external oracles, no aggregation, no spoofing. You can trade anywhere, but only one price matters, on **PulseX**.

**One Price. One Pool. One Truth.**

This is how we kill manipulation.

This is how we protect the protocol.

This is how we win.

# Perpetual Bitcoin